Research Project Periodic Log Entry Form

01st -15th April

Project ID : R26-IT-118

Student ID : IT22279620

Name : Gunasekara G.N.D

Individual Component : Tea-Specific Weather and Climate Intelligence System

Summary of RP work carried out during the reporting period

Date	RP work
1 April	Had a group meeting and discussed the implementation phase of the climate and weather component after completing the proposal.
3 April	Explored available weather and climate datasets from online sources. Found some relevant data but identified limitations in location-specific accuracy.
4 April	Collected and reviewed climate data such as temperature, rainfall, and humidity relevant to tea plantation regions.
7 April	Performed basic preprocessing on the collected data including cleaning, formatting, and organizing datasets.
9 April	Explored suitable models and techniques for weather prediction and analysis.
11 April	Tested initial data with simple model setup and discussed possible improvements with the group.

Supervisor's Comments

- Student progressing well in research project : YES / NO
  If NO
  Project work to be carried out before next evaluation period:
- Code base checked : YES / NO

Signature : .....

Date : 20/04/2026

Research Project Periodic Log Entry Form

15th-30 April

Project ID : R26-IT-118

Student ID : IT22279620

Name : Gunasekara G.N.D

Individual Component : Tea-Specific Weather and Climate Intelligence System

Summary of RP work carried out during the reporting period

Date	RP work
16 April	Refined collected climate dataset and improved data quality by removing missing and inconsistent values.
18 April	Applied preprocessing techniques such as normalization, formatting, and feature selection for model readiness.
19-20 April	Explored suitable machine learning models for weather prediction including regression and time-series approaches.
22-24 April	Implemented an initial model using sample climate data and tested basic prediction outputs.
26-28 April	Evaluated model results and analyzed prediction accuracy for temperature and rainfall patterns.
29 April	Discussed integration of climate prediction module with overall system and planned improvements for model performance and visualization.

Supervisor's Comments

- Student progressing well in research project : YES / NO
  If NO
  Project work to be carried out before next evaluation period:

- Code base checked : YES / NO

Signature : ..... Date : 08/10/2020

Research Project Periodic Log Entry Form

01st – 15th May

Project ID : R26-IT-118

Student ID : IT22279620

Name : Gunasekara G.N.D

Individual Component : Tea-Specific Weather and Climate Intelligence System

Summary of RP work carried out during the reporting period

Date	RP work
01 May	Integrated weather APIs for tea estate climate monitoring.
03 May	Developed disease risk prediction logic using weather datasets.
06 May	Implemented temperature forecasting and climate analytics system.
09 May	Connected climate intelligence module with backend APIs and dashboard.
12 May	Demonstrated climate intelligence prototype during PP1 presentation and live demo.

Supervisor's Comments

- Student progressing well in research project : YES / NO
  If NO
  Project work to be carried out before next evaluation period:

- Code base checked : YES / NO

Signature : .....

Date : 08/10/2026

Research Project Periodic Log Entry Form

16th – 30th June

Project ID : R26-IT-118

Student ID : IT22279620

Name : Gunasekara G.N.D

Individual Component : Tea-Specific Weather and Climate Intelligence System

Summary of RP work carried out during the reporting period

Date	RP work
16 June	Improved weather prediction accuracy through parameter tuning.
19 June	Developed the Multi-output Random Forest Regressor for disease risk prediction.
22 June	Connected weather prediction outputs with disease risk estimation.
26 June	Tested weather prediction and disease risk modules using validation datasets.
29 June	Updated technical documentation based on implementation progress.

Supervisor's Comments

- Student progressing well in research project : ~~YES~~ / NO
  If NO
  Project work to be carried out before next evaluation period:

- Code base checked : YES / NO

Signature : .....

Date : 08/10/2026

Research Project Periodic Log Entry Form

01st – 15th June

Project ID : R26-IT-118

Student ID : IT22279620

Name : Gunasekara G.N.D

Individual Component : Tea-Specific Weather and Climate Intelligence System

Summary of RP work carried out during the reporting period

Date	RP work
02 June	Collected historical weather data from reliable meteorological sources.
05 June	Cleaned and preprocessed weather datasets for model development.
08 June	Performed feature selection for temperature, rainfall, humidity, and wind speed.
11 June	Developed the initial Gradient Boosting Regressor model for weather prediction.
14 June	Evaluated preliminary weather prediction results and documented observations.

Supervisor's Comments

- Student progressing well in research project : YES / NO
  If NO
  Project work to be carried out before next evaluation period:

- Code base checked : YES / NO

Signature : .....

Date : 08/10/2026

Research Project Periodic Log Entry Form

01st – 15th July

Project ID : R26-IT-118

Student ID : IT22279620

Name : Gunasekara G.N.D

Individual Component : Tea-Specific Weather and Climate Intelligence System

Summary of RP work carried out during the reporting period

Date	RP work
02 July	Improved weather prediction accuracy using additional training data.
05 July	Developed recommendation logic for irrigation and harvesting.
08 July	Added fertilizer application recommendations based on predicted weather.
11 July	Conducted functional testing of the weather intelligence module.
14 July	Fixed identified issues and optimized prediction performance.

Supervisor's Comments

- Student progressing well in research project : YES / NO
  If NO
  Project work to be carried out before next evaluation period:

- Code base checked : YES / NO

Signature : .....

Date : 08/10/2026

Research Project Periodic Log Entry Form

16th – 31st July

Project ID : R26-IT-118

Student ID : IT22279620

Name : Gunasekara G.N.D

Individual Component : Tea-Specific Weather and Climate Intelligence System

Summary of RP work carried out during the reporting period

Date	RP work
16 July	Improved backend API performance for weather prediction requests.
19 July	Integrated the weather intelligence module with the web application.
22 July	Conducted end-to-end testing of prediction and recommendation workflows.
26 July	Optimized response time and prediction accuracy.
30 July	Updated implementation documents and testing reports.

Supervisor's Comments

- Student progressing well in research project : YES / NO
  If NO
  Project work to be carried out before next evaluation period:

- Code base checked : YES / NO

Signature : .....

Date : 08/10/2026

Research Project Periodic Log Entry Form

01st – 15th August

Project ID : R26-IT-118

Student ID : IT22279620

Name : Gunasekara G.N.D

Individual Component : Tea-Specific Weather and Climate Intelligence System

Summary of RP work carried out during the reporting period

Date	RP work
02 August	Refined weather prediction based on validation results.
05 August	Improved disease risk prediction accuracy using additional testing data.
08 August	Enhanced recommendation presentation on the dashboard.
11 August	Conducted validation testing and resolved identified issues.
14 August	Prepared demonstration materials for Progress Presentation II (PP2).

Supervisor's Comments

- Student progressing well in research project : YES / NO
  If NO
  Project work to be carried out before next evaluation period:

- Code base checked : YES / NO

Signature : .....

Date : 08/10/2026

Research Project Periodic Log Entry Form

16th – 31st August

Project ID : R26-IT-118

Student ID : IT22279620

Name : Gunasekara G.N.D

Individual Component : Tea-Specific Weather and Climate Intelligence System

Summary of RP work carried out during the reporting period

Date	RP work
16 August	Enhanced the Weather and Climate Intelligence module based on supervisor feedback.
19 August	Improved integration with the overall TeaGuard AI platform.
22 August	Conducted functional and integration testing of the module.
26 August	Updated technical documentation and presentation slides for PP2.
30 August	Finalized the Weather and Climate Intelligence module for PP2.

Supervisor's Comments

- Student progressing well in research project : ~~YES~~ / NO
  If NO
  Project work to be carried out before next evaluation period:

- Code base checked : YES / NO

Signature : .....

Date : 08/10/2026